Projected ROI & Network Impact

Executive Summary

Standard OSRM routing treats every shipment as an isolated, straight-line trip and, in doing so, misses the single largest driver of delay in the Delhivery network: how congestion at one hub bleeds into every downstream corridor that depends on it. Our graph-based engine replaces this static view with one that learns directly from network structure — hub centrality, dwell time, and corridor-level history — to produce ETAs that are accurate enough to act on, and a routing framework that turns that accuracy into measurable cost savings.

On held-out trips, the predictive engine cuts absolute ETA error by **82.9%** relative to raw OSRM output and lifts the share of shipments landing within the **15% SLA accuracy band** from **4.4%** to **60.3%**. Layered on top of this, the FTL-vs-Carting decision framework has already scored **3,983** corridor × time-of-day combinations covering **102,568** historical trips, and identified **Rs. 66.99 Lakh** in addressable cost-saving potential — with nearly **Rs. 10 Lakh** of that sitting behind just three structurally critical hubs.

ROI Dashboard — Graph Intelligence vs. OSRM Baseline

82.9%
Reduction in ETA error (MAE)
Rs. n/a — 209.0 min → 35.7 min

58.4%
Fewer SLA breaches
95.6% → 39.7% breach rate

Rs. 66.99L
Network-wide cost-saving potential
identified across 3,983 corridor windows

Rs. 9.98L
Recoverable from upgrading
just the top 3 bottleneck hubs

Hard ROI: Translating Model Metrics into Business Value

The table below converts the two headline accuracy metrics — Mean Absolute Error (MAE) and on-time accuracy within the 15% SLA band — into the operational language an ops leader can act on.

pastelLavender!30 **Metric**	**OSRM Baseline**	**Predictive Engine**	**Improvement**
ETA error (MAE)	209.0 min	35.7 min	**−82.9%**
pastelBlue!15 SLA breach rate (>15% error)	95.6%	39.7%	**−58.4% (rel.)**
Mean % error (MAPE)	—	18.6%	n/a
pastelBlue!15 Shipments within SLA	4.4%	60.3%	**13.7×**

Source: `baseline_metrics.json`*, evaluated on 28,373 held-out test trips.*

What this means operationally: for every 100 shipments that today arrive outside their promised window under OSRM-based planning, roughly 58 of them would instead land inside the SLA band under the graph-aware engine, with no change to fleet capacity or route infrastructure — the gain comes purely from knowing *which* hubs and time windows are structurally unreliable before the shipment departs.

The FTL-vs-Carting decision framework converts that same structural awareness into a direct routing-cost lever. Of the 3,983 corridor × time-of-day strata scored, **90.2%** are flagged *chronic* (actual time >120% of OSRM), and switching mode on just the 471 strata (11.8%) where FTL beats Carting on cost recovers **Rs. 16.05 Lakh**; tightening Carting allocation on the remaining strata recovers a further **Rs. 50.94 Lakh** — a combined network opportunity of **Rs. 66.99 Lakh**.

Top 3 bottleneck hubs — where the savings concentrate:

pastelLavender!30 **Hub ID**	**Centrality Tier**	**Chronic Corridors**	**Trips Affected**	**Cost-Saving Potential**
IND000000ACB	Critical	100	16,431	Rs. 4.64L
pastelGreen!15 IND562132AAA	High	91	7,346	Rs. 2.90L
IND421302AAG	Moderate	90	6,369	Rs. 2.44L
pastelGreen!15 **Combined**	—	**281**	**30,146**	**Rs. 9.98L**

Call to Action: Pilot the Worst Corridor Immediately

Recommended Immediate Pilot

The single highest-value corridor in the dataset is **IND160002AAC** → **IND562132AAA** (Evening, 16:00–20:00), a 1,056 km lane currently running on Carting that the framework flags for an immediate switch to **FTL**.

- Average delay factor of **1.66**× OSRM, with excess delay of **625 minutes** per trip versus the OSRM prediction.
- ₹ **Rs. 19,973** in cost saving identified for this corridor-window alone (81 trips), the largest single recommendation in the network.
- Source hub sits in the **High** centrality tier — delays here propagate downstream, inflating the true cost of inaction beyond this lane.
- Confidence rating: **HIGH**, with no chokepoint override required.

Recommendation: authorise a two-week mode-switch pilot on this corridor immediately. Track realised cost-per-shipment and on-time performance against the model's projection before extending the framework to the remaining 470 FTL-flagged corridors.

Figures computed directly from the decision framework output (`ftl_carting_recommendations.csv`, 3,983 corridor × time-of-day rows, 102,568 trips) and `baseline_metrics.json` (XGBoost baseline vs. OSRM, 28,373 held-out test trips). No raw model internals are reproduced here; all figures are decision-ready business metrics.